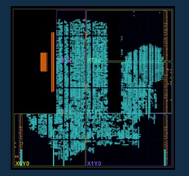
SoundSight

3D Sound Localization and Speaker-Gated Pan–Tilt Tracking

Zynq-7020
PL + PS / PYNQ



Actual FPGA implementation

RGB touchscreen **Directional audio** **3D localization** **Speaker verification** **Pan–tilt tracking**

INTERNAL DRAFT · 2026-10-31

Affiliation, team credits and print size to confirm

01 Application value

SoundSight combines sound direction, speaker verification and pan–tilt control for local acoustic interaction. Localization estimates azimuth and elevation, not source distance.

Meetings and classrooms

Directional audio could reduce manual aiming as speakers move or take turns.

Robots and interactive devices

Direction estimates support turning toward a voice. Speaker-gated mode adds authorization for an enrolled user.

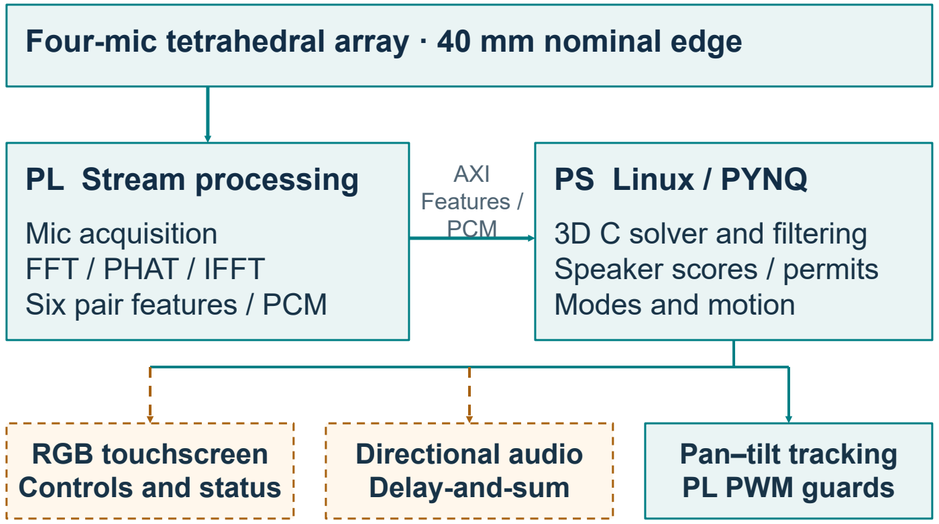
Standalone RGB touchscreen

FPC display and touch for modes, settings and status. Integration awaits evidence.

Device and touchscreen photos
Original front / 45° views and interface needed

Benefits depend on calibration, reliable verification and sustained operation.

02 System and design



Functional diagram. Dashed paths await integration validation. Speaker input is currently single-mic PCM, not verified DAS output.

Core design contributions

Decoupled acquisition and processing
PL handles streams; PS runs estimation, verification and UI.

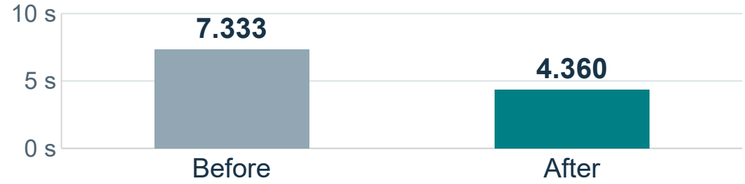
Bounded queues and fresh results
Drop old READY results, protect LOCKED frames and reject stale angles.

Layered motion permission
Separate tracking modes; axes default off. Direction and STOP guards remain.

03 Key results

Offline speech front-end optimization

40.55% less processing time



Two 10.912 s recordings, sequential board benchmark. Loading excluded. Scores unchanged; inference time essentially unchanged. Not an online end-to-end gain. [E1]

Speaker-gated pan–tilt field tests

Metric	Run 1	Run 2
Duration / s	32.159	27.254
Frames consumed	3012	2552
Motion commands	117	94
Runtime fault value	0	0
Old results dropped	0	1

Trial threshold 0.45; permit 10 s. User observed tracking. Commands do not prove arrival; Run 2 has one sequence gap. Short trials do not establish accuracy or long-term stability. [E2]

Next: measure 3D error, identity rejection and directional audio.

A time-limited permit does not bind a person to a direction in a multi-speaker scene. CAM++, Silero and vendor IP retain their attribution. Sources: notes.